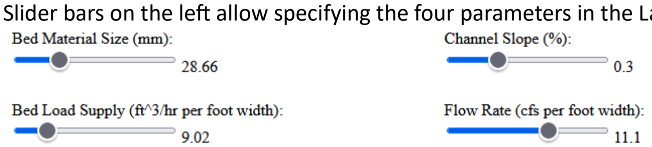
Notes on the Lanes Balance Scale Application

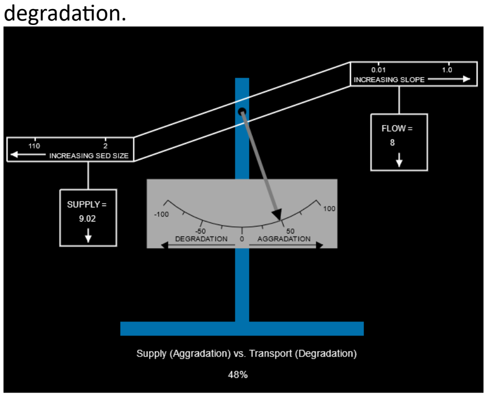
1. Lane's Balance Equation:
   Sediment Supply (Qs) * Sediment Size (d50) ~ Flow Rate (Q) * Slope (S)
   Sediment <u>Supply</u> (left side) ~ Sediment <u>Transport</u> Capacity (right side)

2. Slider bars on the left allow specifying the four parameters in the Lane's Balance Equation:

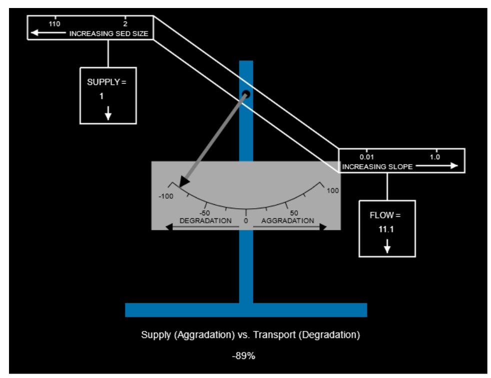


3. These specified values are used to compute the percent balance for Supply (Aggradation) vs. Transport (Degradation). The balance computation is based on comparison with an assumed stable condition.

4. The Lane's Balance Scale Graphic on the top left shows the specified parameters on the balance scale bar and the computed percent balance on the scale indicator. Positive values indicate aggradation while negative values indicate degradation.

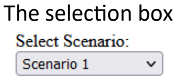


Supply side is heavier → Aggradation
(Created with Size=28.66, Supply=9.02, Slope=0.3, Flow=8)



Transport side is heavier → Degradation
(Created with Size=28.66, Supply=1.0, Slope=0.3, Flow=11.1)

5. The selection box on the right allows selecting one of three items for investigation of stream stability scenarios.



6. Two slider bars and four command buttons are present in the lower right corner. These input elements enable the following functionality:

Input Element:	Function:
"Default Watershed"	Sets the stable condition and current parameter values based on the default watershed values for Area (15 sq.mi.) and Slope (0.3%). Note: This button resets the parameters to the original stable conditions used for the scenarios described in #5.
"Set Watershed"	Sets the stable condition and current parameter values based on the specified slider values for watershed area and watershed slope.
"Set Stable"	Sets the stable condition based on the currently specified slider bar values for size, supply, slope (left side), and flow.
"Reset Stable"	Resets the slider bar values to the default stable condition values (set initally or by one of the 3 command buttons above).